

Problem Set 1 — Kinematics

VCE Specialist Mathematics 3&4 · Chapter 12 · Week 1

Due: start of Week 2.

Name: _____

Due: Monday 20 July

Section A — Position, velocity, acceleration

A1. A particle moves in a straight line with position $x(t) = 2t^3 - 9t^2 + 12t$ metres at time t seconds, $t \geq 0$.

- a.) Find $v(t)$ and $a(t)$. (2 marks)
- b.) Find the times at which the particle is at rest. (2 marks)
- c.) Find the acceleration at each of these times. (2 marks)

A2. For the particle in A1, find the *displacement* and the *distance travelled* during the first 3 seconds. (4 marks)

A3. Hence write down the average velocity and the average speed over the first 3 seconds. (2 marks)

Section B — Constant acceleration and v - t graphs

B1. A car travelling at 30 m/s brakes with constant deceleration 5 m/s².

- a.) How long does it take to stop? (1 marks)
- b.) How far does it travel while stopping? (2 marks)

B2. A train accelerates uniformly from rest at 0.5 m/s² for 40 s, travels at constant velocity for 60 s, then decelerates uniformly to rest at 1 m/s².

- a.) Sketch the velocity–time graph. (2 marks)
- b.) Find the total distance travelled. (3 marks)
- c.) Find the average speed for the whole journey. (1 marks)

B3. A particle has velocity $v(t) = 12 - 3t$ m/s for $0 \leq t \leq 6$.

- a.) Find the displacement over $[0, 6]$. (2 marks)
- b.) Find the distance travelled over $[0, 6]$. (3 marks)

Section C — Acceleration as a function of v or x

C1. A body moving through mud decelerates according to $a = -0.2v$ (m/s²), with initial velocity 40 m/s at $x = 0$, $t = 0$.

- a.) Using $a = \frac{dv}{dt}$, find v as a function of t . (3 marks)
- b.) Find the time taken for the velocity to fall to 10 m/s. (2 marks)
- c.) Using $a = v \frac{dv}{dx}$, find v as a function of x , and hence the distance travelled before the body comes to rest. (3 marks)

C2. A particle moves so that $a = -4x$, and $v = 6$ m/s when $x = 0$.

- a.) Using $a = \frac{d}{dx}(\frac{1}{2}v^2)$, show that $v^2 = 36 - 4x^2$. (3 marks)
- b.) State the region in which the motion is confined, and find the speed at $x = 2$. (3 marks)

C3. A skydiver of unit mass falls so that $a = 9.8 - 0.1v^2$ (downwards positive).

- a.) Explain why the velocity approaches a limiting (terminal) value, and find it. (3 marks)
- b.) Write (do not solve) the differential equation whose solution gives v as a function of the distance fallen. (2 marks)

Answers

A1. a.) $v = 6t^2 - 18t + 12 = 6(t-1)(t-2)$; $a = 12t - 18$. b.) $t = 1, 2$ s. c.) $a(1) = -6$, $a(2) = 6 \text{ m/s}^2$.

A2. $x(0) = 0$, $x(1) = 5$, $x(2) = 4$, $x(3) = 9$. Displacement = 9 m; distance = $5 + 1 + 5 = 11$ m.

A3. Average velocity = 3 m/s; average speed = $\frac{11}{3} \approx 3.67$ m/s.

B1. a.) 6 s. b.) 90 m.

B2. b.) Phases: $400 \text{ m} + 1200 \text{ m} + 200 \text{ m} = 1800 \text{ m}$ (deceleration takes 20 s). c.) $1800/120 = 15 \text{ m/s}$.

B3. a.) $\int_0^6 v \, dt = 18 \text{ m}$. b.) $v = 0$ at $t = 4$: $\int_0^4 = 24$, $\int_4^6 = -6$; distance = 30 m.

C1. a.) $v = 40e^{-0.2t}$. b.) $t = 5 \ln 4 \approx 6.93$ s. c.) $v \frac{dv}{dx} = -0.2v \Rightarrow \frac{dv}{dx} = -0.2 \Rightarrow v = 40 - 0.2x$; rest at $x = 200$ m.

C2. a.) $\frac{1}{2}v^2 = -2x^2 + 18$ using $v(0) = 6$. b.) $v^2 \geq 0 \Rightarrow -3 \leq x \leq 3$; at $x = 2$, $v^2 = 20$, speed = $2\sqrt{5} \approx 4.47 \text{ m/s}$.

C3. a.) As v grows, $a \rightarrow 0$; terminal velocity where $9.8 = 0.1v^2$: $v = \sqrt{98} = 7\sqrt{2} \approx 9.90 \text{ m/s}$. b.) $v \frac{dv}{dx} = 9.8 - 0.1v^2$.